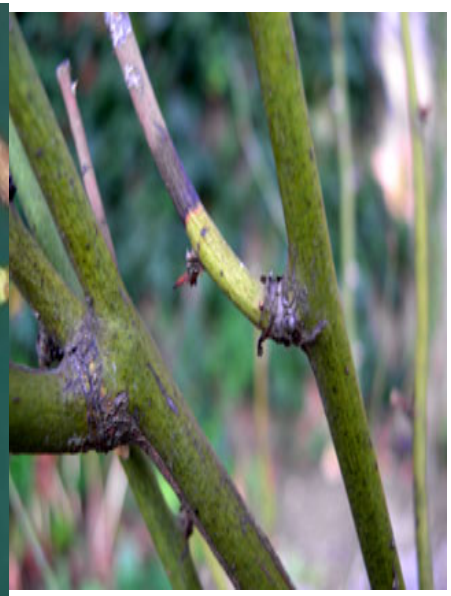
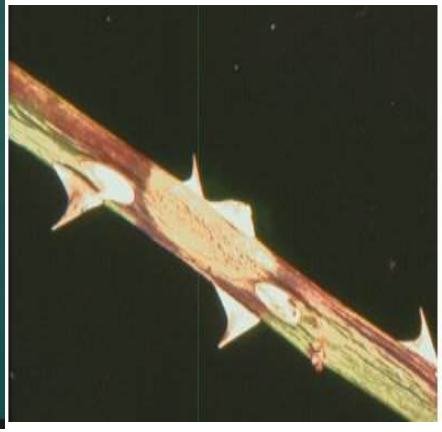
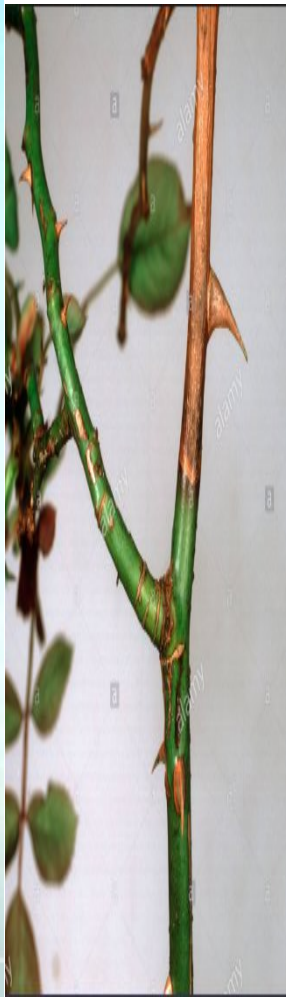
Diseases of rose

1. Die back - *Diplodia rosarum*, *Botryodiplodia theobromae*
2. Verticillium wilt - *Verticillium albo-atrum, V.dahliae*
3. Stem canker - *Coniothyrium fuckelii*
   Common canker PS - *Leptosphaeria coniothyrium*
4. Black spot - *Diplocarpon rosae* (IPS: *Marssonina rosae*)
5. Powdery mildew - *Sphaerotheca pannosa var. roseae*
6. Alternaria leaf spot - *Alternaria alternata, A. tenuis* var *rosicola*
7. Rust - *Phragmedium mucronatum, P.fusiformie, P.tuberculatum, P. rosae-pimpinelifoliae*
8. Crown gall - *Agrobacterium tumefaciens*

Die back - *Diplodia rosarum, Botryodiplodia theobromae*







Die back - *Diplodia rosarum, Botryodiplodia theobromae*

Symptoms

- Die-back generally occurs after pruning.
- The infection occurs at the pruned ends of stems or through wounds caused due to harvesting of flowers or other operations.
- Infection spreads from twig to main stem and roots and finally whole plant is killed
- Damage caused by **digger wasps** also facilitates the infection.
- **Old plants and neglected bushes are more prone** to die-back.

Die back - *Diplodia rosarum, Botryodiplodia theobromae*

Symptoms contd…..

- The infected portion become brown to dark brown or black
- Browning and rotting of infected stem extends inside the stem and killing the entire plant.
- The internal spread of the disease in the form of brown discoloration can be observed when affected stems are split open.
- The fruiting bodies of the fungus are seen as small eruptions around the bases of thorns on the infected stems.

Die back - *Diplodia rosarum, Botryodiplodia theobromae*

Survival and spread

PSI : Pycnidia present in dead twigs

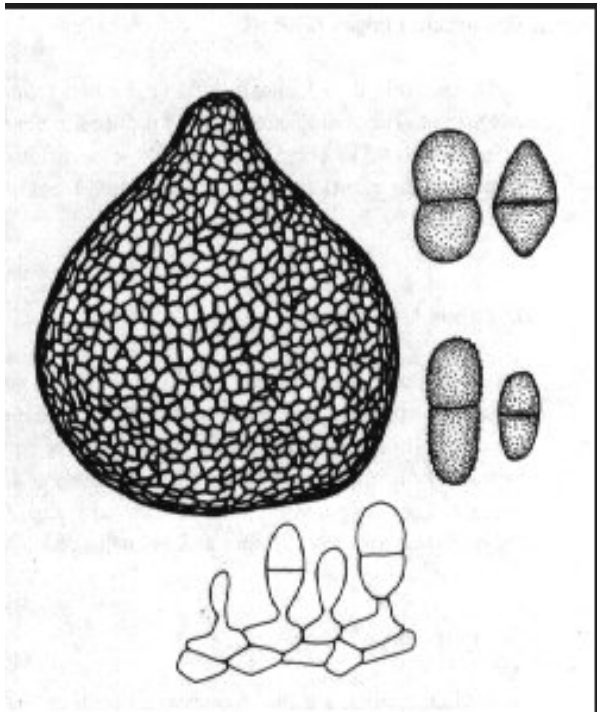
SSI: Conidia through irrigation water and wind

- The fungus ***Alternaria alternata*** is also reported to cause die-back in *Rosa damascene*.
- Plants defoliated by **black spot and Powdery mildew diseases** are more susceptible because of depletion of carbohydrates in the stems

Die back - *Diplodia rosarum, Botryodiplodia theobromae*

The fungus produces round, black pycnidia which bear spores.

The pycnidiospores are dark coloured and two celled.



Die back - *Diplodia rosarum, Botryodiplodia theobromae*

Management

- Diseased twigs should be removed and destroyed
- The **secature should be disinfected**
- While **pruning, a slant cut is advised**
- After pruning the cut end should be protected with Chaubattia paste (4 parts of copper carbonate, 4 parts of red lead and 5 parts of linseed oil with traces of BHC to ward off the digger wasp)

Die back - *Diplodia rosarum, Botryodiplodia theobromae*

Management contd…

- Resistant varieties: White Sign, Red Gold, Bhim, Summer Queen, Quebec, Resolute, Samba and Blue Moon are resistant.
- Spray difoltan 0.2% or mancozeb 0.2% or chlorothalonil in early September and late October
- Systemic fungicides like carbendazim may be used as soil drench.

Verticillium wilt - *Verticillium albo-atrum, V.dahliae*

Verticillium wilt - *Verticillium albo-atrum, V.dahliae*

Wilt is serious in the nursery.

Symptoms

- **Rooted cuttings die after sprouting**
- The entire plant shows light yellowing followed by drying resulting the death of the plant

Verticillium wilt - *Verticillium albo-atrum, V.dahliae*

Disease cycle & Management

Refer to earlier notes on wilt

Stem canker/ Common canker - *Coniothyrium fuckelii* PS - *Leptosphaeria coniothyrium*

- **Warm moist propagation conditions** are ideal for occurrence and spread of the pathogen.
- Plants under **storage, transit or under stress** are highly susceptible to the disease.

Stem canker/ Common canker - *Coniothyrium fuckelii* PS - *Leptosphaeria coniothyrium*

Small spots on the bark that gradually expand to girdle the stem.

Dried tissue with numerous fruiting bodies inside the cankerous region.

Stem canker/ Common canker - *Coniothyrium fuckelii* PS - *Leptosphaeria coniothyrium*

Stem canker/ Common canker - *Coniothyrium fuckelii* PS - *Leptosphaeria coniothyrium*

Stem canker/ Common canker - *Coniothyrium fuckelii* PS - *Leptosphaeria coniothyrium*

Most common in Chandigarh.

Symptoms

- **Pathogen can affect any part of the plant**, but most common on stems and canes
- Produce **red or yellow** spot on the surface of the cane.
- The affected areas gradually enlarge and become distinctly **brown or reddish- brown in the centres** as they increase in size
- The surface of the spot becomes **slightly wrinkled or wart like** over the submerged fruiting structures (pycnidia) of the fungus

Stem canker/ Common canker - *Coniothyrium fuckelii* PS - *Leptosphaeria coniothyrium*

Symptoms contd….

- **Stems may be girdled** with die back symptoms
- Cankers **girdle** or encircle a cane or stem result in **wilting and death** of cane or stem above the canker.
- Canker initiates around **insect punctures, thorn picks or abrasions caused by tying & sun scald injury.**
- Small, **black flecks** can develop in the centres of the cankers

Stem canker/ Common canker - *Coniothyrium fuckelii* PS
- *Leptosphaeria coniothyrium*

Pathogen

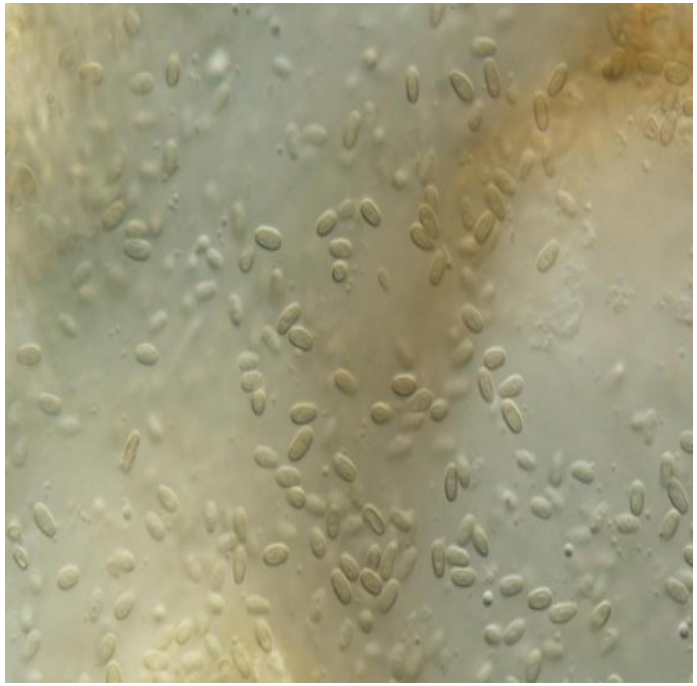
- Two fungi *Coniothyrium wernsdorffiae* which cause brand canker and *Coniothyrium fuckelii* cause common canker

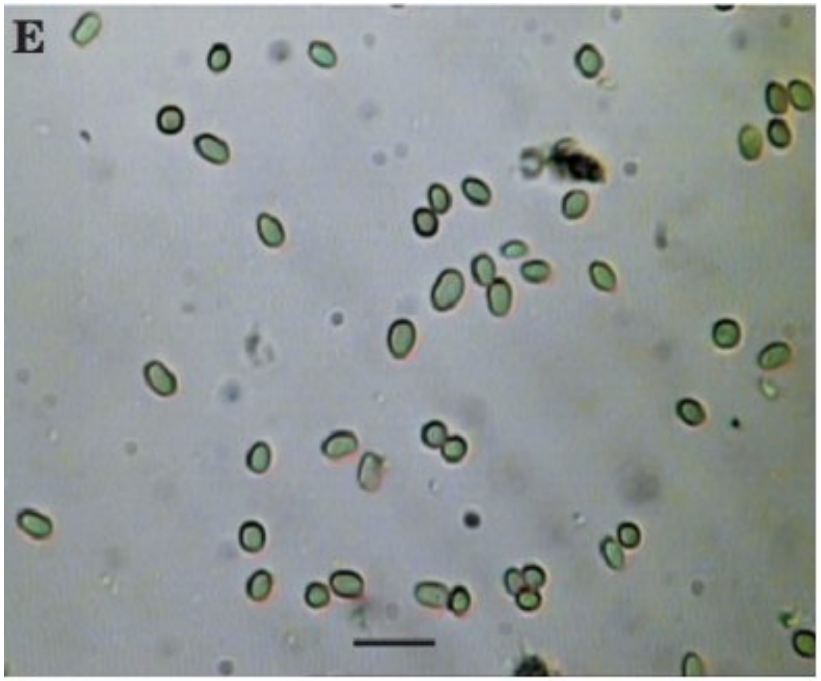
- **Both fungi infect through wounds from pruning, insects, and other types of mechanical injuries.**

Survival and spread

- Pycnidia in infected twigs
- Pycnidiospores

Stem canker/ Common canker - *Coniothyrium fuckelii* PS - *Leptosphaeria coniothyrium*





- Rose canker (PS*: Leptosphaeria coniothyrium*).
- Pycnidia uni-loculate, ostiolate and globose to subglobose, conidia hyaline to light brown, smooth, aseptate, spherical to subspherical, ellipsoidal

Stem canker/ Common canker - *Coniothyrium fuckelii*) PS - *Leptosphaeria coniothyrium*

Management

- The **cut should be just above the bud or leaf axil** while pruning or removing flowers & dried blooms.
- Spraying Mancozeb .2% is recommended.

Black spot - *Diplocarpon rosae* (IPS: *Marssonina rosae)*

Black spots with diffuse margins.

Numerous black spots can be seen on the leaves.

Black spot - *Diplocarpon rosae* (IPS: *Marssonina rosae)*

Large spots or lesions with a feathery edge (not smooth).

Numerous spots unevenly distributed **along the main veins** of the leaf.

Black spot - *Diplocarpon rosae* (IPS: *Marssonina rosae)*

Leaf and stem lesions with lots of defoliation.

Heavily defoliated bush with very few leaves left

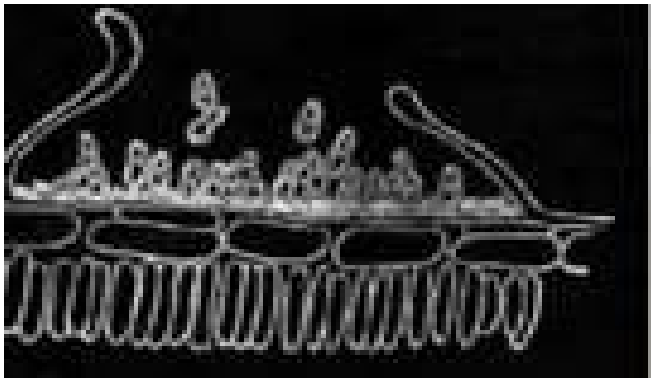
Black spot - *Diplocarpon rosae* (IPS: *Marssonina rosae)*

Black spot - *Diplocarpon rosae* (IPS: *Marssonina rosae)*

Black spot - *Diplocarpon rosae* (IPS: *Marssonina rosae)*

- Acervuli are formed on leaves and young shoots during the summer, while apothecia are borne on decaying overwintering leaves.
- Conidia are hyaline, oval to elliptical with single septation and are formed in an acervulus.

Fig. 11. Acervuli of Diplocarpon rosae, the cause of black spot, in a lesion with feathery margins.



Black spot - *Diplocarpon rosae* (IPS: *Marssonina rosae)*

- It has also been called as **leaf blotch, leaf spot, blotch, rose actinonema and rose leaf asteroma and star sooty mould.**

- This disease is considered as most severe disease of garden roses worldwide including India.

Symptoms:

- Brown to black, nearly circular spots (2-12 mm in dia.) on both sides of leaves often has **fringed margins with a feathery hyphal growth**

- The affected leaves turn yellow and drop prematurely due to the formation of ethylene which is more in green leaves infected with this disease.

Black spot - *Diplocarpon rosae* (IPS: *Marssonina rosae)*

Symptoms contd…..

- Other above ground parts like **stipule, petiole, peduncle and sepals** may also develop similar spots.
- **Infections on petiole and stipule** are usually inconspicuous and small **while petal infection usually causes formation of small, reddish spots** accompanied by distortion of the surrounding tissues.
- Close examination of the lesion reveal the presence of small black acervuli which may be distributed or concentric circles over lesion surface.

Black spot - *Diplocarpon rosae* (IPS: *Marssonina rosae)*

Pathogen:

- *Diplocarpon rosae* (IPS: *Marssonina rosae*)
- The **apothecia are uncommon**, globose to disc shaped, subcuticular, radiate and are 100-300 µm in dia.
- Asci are inoperculate, oblong, and cylindrical, short stalked, 70-80 x 12-18 µm and contain a pore.
- Each ascus contains 8 ascospores which are oblong, elliptical, hyaline, and unequally two-celled, constricted at the septum and are 20-25 x 5-6 µm in size.

Black spot - *Diplocarpon rosae* (IPS: *Marssonina rosae)*

Pathogen contd…

- The paraphyses are numerous, filiform, septate and enlarged at the tips.
- Conidia are 18-25 x 5-6 µm in size, hyaline, oval to elliptical with single septation and are formed in an acervulus.
- Acervuli are formed on leaves and young shoots during the summer, while apothecia and spermogonia are borne on decaying overwintering leaves.

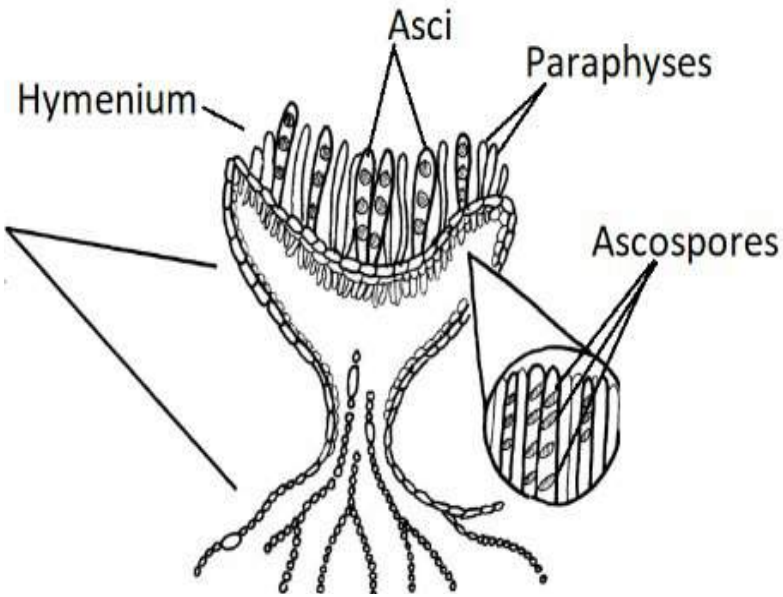
Black spot - *Diplocarpon rosae* (IPS: *Marssonina rosae)*

•The apothecia are globose to disc shaped, subcuticular, radiate.

• Asci are inoperculate, oblong, and cylindrical, short stalked, and contain a pore.

•Each ascus contains 8 ascospores which are oblong, elliptical, hyaline, and unequally two-celled, constricted at the septum .

•The paraphyses are numerous, filiform, septate and enlarged at the tips



Black spot - *Diplocarpon rosae* (IPS: *Marssonina rosae)*

Disease cycle and epidemiology:

- The fungus perennates on **overwintering rose leaves and infected stem tissue.**
- Although the perfect stage has been recorded in certain geographic areas, but their role is not understood in the pathogen life cycle.
- For rapid spread of the disease, temperature ranging between 0-33°C, with optimum temperature (24°C) and more than 85% relative humidity is favourable.

Black spot - *Diplocarpon rosae* (IPS: *Marssonina rosae)*

Management:

- Removal of **overwintering inoculum by eliminating** infected leaves is helpful in limiting the inoculum spread.
- **Avoid overhead irrigation** and dense planting to allow good air circulation through leaf canopy and **do not allow leaves to remain wet for more than 7- 12 hours**
- Sprays of Bordeaux mixture, wettable sulphur, carbamates, ergosterol-inhibiting fungicides like triforine, benomyl-zinc, chlorothalonil, mancozeb provide effective control of this disease.
- **Resistant varieties Jawahar and Pusa Sonia**

Powdery mildew - *Sphaerotheca pannosa* var. *roseae*

- Theophrastus gave the first account of powdery mildew on roses around 300 BC, but Wallroth in 1819 first described the fungus causing this disease as *Alphitomorpha pannosa*.

- It was transferred to the genus *Erysiphe* as *E. pannosa* (Walk.) Fr. in 1829 and finally was described and placed in the genus *Sphaerotheca* in 1851.

Powdery mildew - *Sphaerotheca pannosa* var. *roseae*

Powdery mildew - *Sphaerotheca pannosa* var. *roseae*

Powdery Mildew (*Podosphaera pannosa*)

Powdery mildew - *Sphaerotheca pannosa* var. *roseae*

Powdery mildew - *Sphaerotheca pannosa* var. *roseae*

- This is **a very serious disease in open and protected structures** and affects both quality and quantity of cut flowers.

- Severe mildew growth reduces leaf growth, the aesthetic value of plants, photosynthetic efficiency and there by plants growth and stability of cut flowers.

Symptoms:

- **Small, discrete lesions appear on the stems, leaves** or sometimes on the flowers

- Production of numerous conidia gives the affected plants a white powdery appearance.

Powdery mildew - *Sphaerotheca pannosa* var. *roseae*

Symptoms contd…

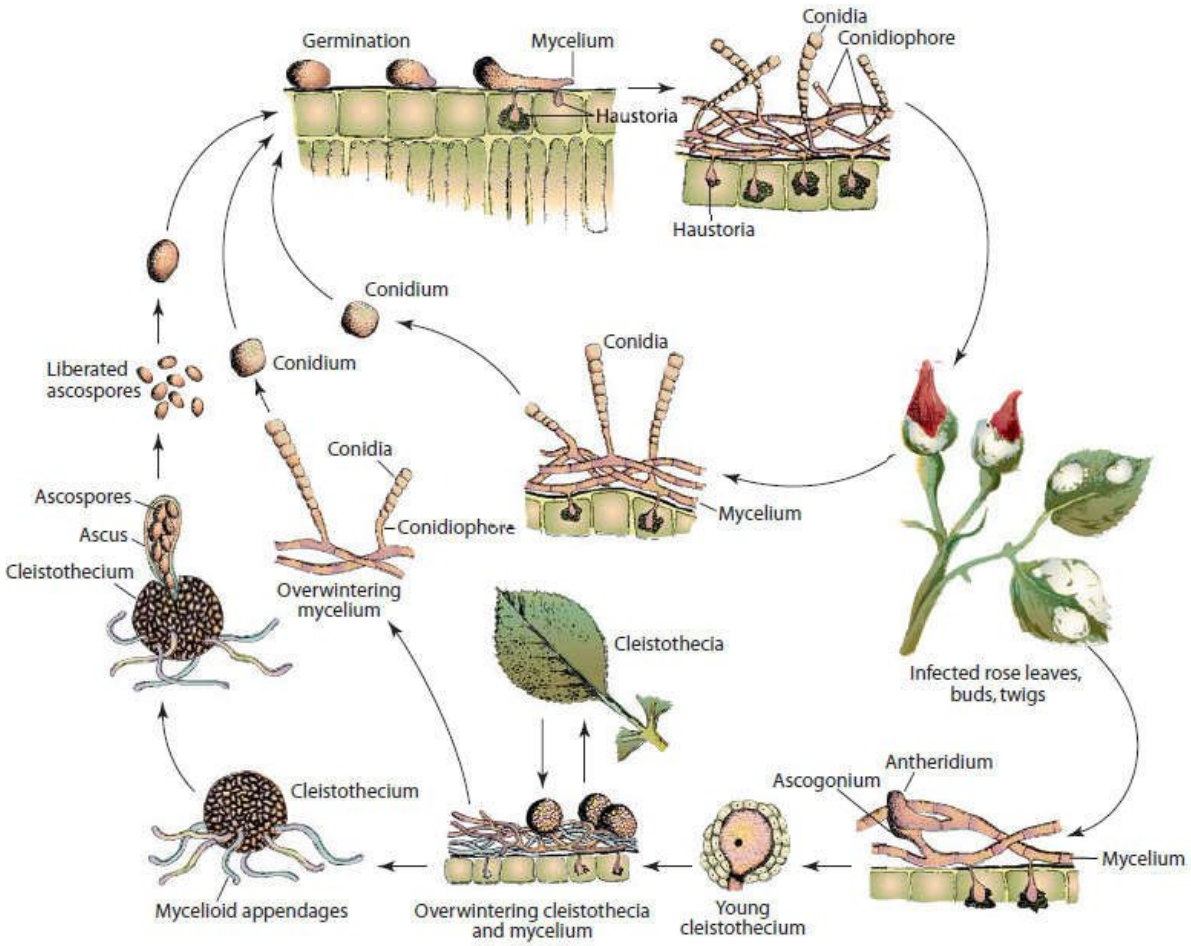
- **Newly unfolded leaves are most susceptible**, becoming increasingly resistant with age.
- The young growing tips are completely covered by the mildew.
- As the cells are killed, dark spots may be noticed on the infected leaves.
- Infected flower buds do not open.
- The petals may become discoloured, dwarfed and finally die.

Powdery mildew - *Sphaerotheca pannosa* var. *roseae*

Pathogen:

- The disease is caused by *Sphaerotheca pannosa* var. *roseae*
- The appendages on cleistothecia of *S. pannosa* **are vestigial or lacking and its ascocarps are usually embedded in felt like mycelium.**
- Each cleistothecium contain single ascus which bear usually 8 hyaline and elliptical ascospores.

Powdery mildew - *Sphaerotheca pannosa* var. *roseae*



Powdery mildew - *Sphaerotheca pannosa* var. *roseae*

Disease cycle and epidemiology:

- **Infection of dormant buds is** the primary mode of perennation.
- Conidia are produced at the primary infection sites readily dispersed by air movement.
- Warm and humid weather with cool nights favours the development of the disease.
- A temperature of 21°C and relative humidity ranging between 97-99% is quite effective for the spread of the pathogen.

Powdery mildew - *Sphaerotheca pannosa* var. *roseae*

Management:

- **Pruning of infected shoots** at the end of the season and burning of these shoots will help **to prevent over wintering of the fungus.**
- In **greenhouse roses lowering night humidity** by fans / venting / heating.
- Spraying the crop with fungicides like hexaconazole (0.05%), difenoconazole (0.03%), triadimefon (0.075%), fenarimol (0.04%) or carbendazim (0.1%) effectively check this disease.

Powdery mildew - *Sphaerotheca pannosa* var. *roseae*

Management contd…

- Biological control of pathogenic fungus can be obtained by applying hyperparasites like *Penicillium* sp., *Cladosporium* sp., *Tilletiopsis pallescens* and *Sporothrix flocculosa* (yeast-like phylloplane fungi) ,

Alternaria leaf spot - *Alternaria alternata*, *A. tenuis* var *rosicola*

Alternaria leaf spot - *Alternaria alternata*, *A. tenuis* var *rosicola*

Symptoms

- The disease was common occurrence during **the rainy and winter seasons** (June to December) but disappear in summer season
- Small oval to irregular, dull brown to **black spots are first noticed on the margin and apex of leaves**
- Initially the spots are **scattered but later enlarge** become confluent and cover the entire leaf surface.
- Sometimes the **individual spots show concentric rings**, infected leaves become brittle.
- Defoliation is quite common up to 60%
- Under humid conditions, **flower buds and flowers are also infected.**
- The **stem and twigs are invariably free.**

Rust - *Phragmedium mucronatum, P.fusiformie, P.tuberculatum, P. rosae-pimpinelifoliae*

Fig. 18. Chlorotic spots of rust, caused by Phragmidium mucronatum, on upper leaf surfaces. (Courtesy K. Ohkawa)

Fig. 17. A close-up of the uredial stage of the rust fungus Phragmidium mucronatum on a lower leaf surface. (Reprinted, by permission, from Horst, 2001)

Rust - *Phragmedium mucronatum, P.fusiformie, P.tuberculatum, P. rosae-pimpinelifoliae*

Fig. 22. The telial stage and the uredial stage of the rust fungus Phragmidium mucronatum on the same leaf. (Courtesy K. Milne)

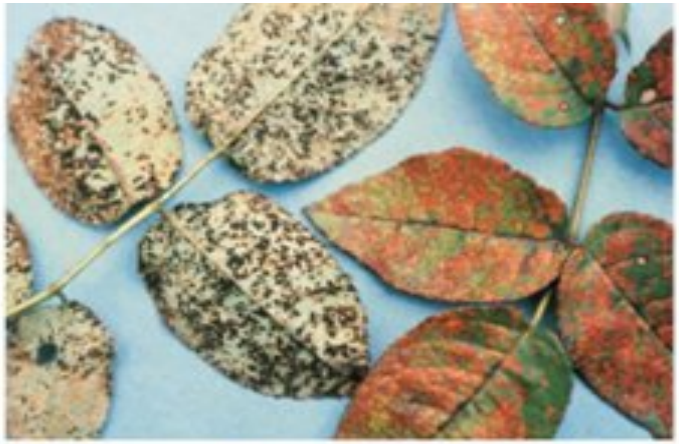


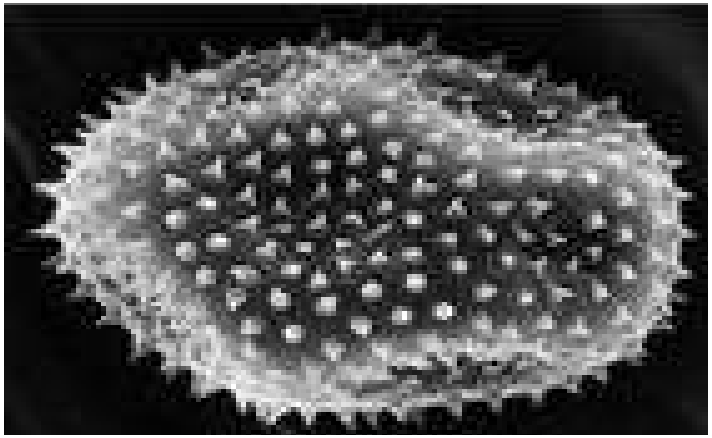
Fig. 20. Black pustules of the telial stage and orange pustules of the uredial stage of the rust fungus Phragmidium mucronatum on leaf surfaces. (Courtesy V. A. Wager)

Fig. 23. The overwintering telial stage of the rust fungus Phragmidium mucronatum on canes.

Rust - *Phragmedium mucronatum, P.fusiformie, P.tuberculatum, P. rosae-pimpinelifoliae*

Rust - *Phragmedium mucronatum, P.fusiformie, P.tuberculatum, P. rosae-pimpinelifoliae*

Aeciospores



Uredospores

Teliospore of *Phragmidium rosae-rugosae* on *Rosa rugosa* (x20)

Rust - *Phragmedium mucronatum, P.fusiformie, P.tuberculatum, P. rosae-pimpinelifoliae*

- The disease is of **worldwide distribution**. Heavy losses occur due to premature defoliation.

Symptoms

- On upper leaf surface**, minute, orange pycnia** appear.
- **Yellow to brown pustules impart rusty appearance** to the shoots.
- The fungus produces **aecial stage in spring**.
- The **aecia are prominent as yellowish lesions on the lower surface** and the spores get released by the rupture of the epidermis .

Rust - *Phragmedium mucronatum, P.fusiformie, P.tuberculatum, P. rosae-pimpinelifoliae*

Symptoms contd…

- They are erumpent, bright orange, usually surrounded by a distinct but narrow chlorotic area.
- **Aecia and pycnia also appear on petioles and young woody stems.**
- After aeciospores infect the leaves, **the uredinial stage starts forming as a whitish fleck on the lower surface of the lesion.**
- **Mature uredinia are reddish, orbicular-shaped and contain orange coloured urediniospores** which are released 14 days after the leaf become infected
- These spores continue to cause new infections throughout the summer.

Rust - *Phragmedium mucronatum, P.fusiformie, P.tuberculatum, P. rosae-pimpinelifoliae*

Pathogen:

- **Several species of *Phragmedium*** are responsible for this diseases but *P. mucronatum*, *P.fusiformie*, *P.tuberculatum*, *P. rosae-pimpinelifoliae* are important.

Disease cycle and epidemiology:

- The rust fungus overwinters either in the form of **teliospores or as mycelium** in the infected stems.
- The optimum temperature range for aeciospores and urediniospores germination lies in between 15 to 21° C while teliospores are produced at 18° C.
- Wind helps in release of rust spores and their short and long distance dispersal.

Rust - *Phragmedium mucronatum, P.fusiformie, P.tuberculatum, P. rosae-pimpinelifoliae*

Management:

- **Remove infected leaves and stem infection** near the crown and destroy them.
- Severe pruning to be taken up to destroy the stem infection.
- Spray the plants with dithiocarbamates, sulphur fungicides, oxycarboxin or ergosterol biosynthesis inhibiting fungicides.
- Hyperparasites such as ***Darluca filum*, *Tuberculina vinosa* and *Verticillium lecani*** can be used as biocontrol agents

Crown gall - *Agrobacterium tumefaciens*

Crown gall - *Agrobacterium tumefaciens*

Galls occur near soil line as well as near pruning cuts as in this image.

Galls have started at pruning wounds. Also, the cross-section of the gall on the right shows disorganized but solid tissue.

Crown gall - *Agrobacterium tumefaciens*

Galls may be numerous on canes or branches.

Galls can be seen above or below ground.

Crown gall - *Agrobacterium tumefaciens*

Most of the time galls appear below ground and may be overlooked as a cause for poor growth

Crown gall - *Agrobacterium tumefaciens*

Symptoms:

- The disease is initiated with formation of galls on the stems and roots.
- Galls first appear as small protuberances on the plant surface and may eventually enlarge to 10 cm or more in size and are commonly found on crown portion
- These are soft at first, but later may become hard and woody.
- Plant growth slows down and in succeeding year many plants die.
- Sometimes aerial galls are produced on the stem, leaf petioles and cut ends of stem where flowers had been removed

Crown gall - *Agrobacterium tumefaciens*

Pathogen:

- The bacterium is aerobic, gram negative, rod shaped and non-sporulating measuring 0.4-0.8 x 1.0 -3.0 µm.
- They are motile and have 1 to 5 lateral flagella, but non-motile variants are also in existence.
- The colonies are convex, circular, glistening and translucent.

Crown gall - *Agrobacterium tumefaciens*

Disease cycle and Epidemiology:

- The bacterium persists saprophytically in the soil for long periods in the absence of the host plants.
- Latent infections can also be a means of dissemination.
- It can multiply and spread rapidly at an optimum temperature of 25-30°C.
- The bacterium enters the plant through wounds produced after pruning or during cultivation or insect damage.
- Disease spread through infected plants, movement of contaminated soil, water, tools etc.
- Overhead sprinklers and insects are major causes of extensive aerial gall formation.

Crown gall - *Agrobacterium tumefaciens*

Management:

- Integration of practices like chemical treatment, selection of clean stock, sanitation, biocontrol methods judiciously can reduce disease.
- Heating of soil to a temperature of 60°C for 30 minutes in glasshouse by following
- Steaming or aerated steam is useful.
- Tools used for pruning, cuttings or cultivation should be thoroughly washed and disinfected at frequent intervals and care should be taken to minimize wounding during cultivation.

Crown gall - *Agrobacterium tumefaciens*

Management contd…

- Gallex or Bacticin (0.1%), copper oxychloride (0.3%), copper sulphate (0.2%) can be applied on wounded areas.
- Biocontrol agent *Agrobacterium radiobacter* (K-84), a non pathogenic strain can be applied as seed or pre-plant dip for minimizing the disease spread.